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Empirical Project

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Impact of Industrial Robots on U.S. Employment and Wages

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Abstract

This study aims to replicate the paper *Robots and Jobs: Evidence from US Labor Markets* by Acemoglu and Restrepo (2017), which examines the impact of industrial robots on labor markets in the United States and to offer a complementary analysis to refine the understanding of the impacts of robotization on employment and wages. Using the same methods and datasets, the study assesses the effect of robots on employment and wages in U.S. commuting zones. The results show a positive association between robot exposure and the number of integrators, as well as a significant negative impact on employment and wages. An increase of one robot per thousand workers reduces the employment-to-population ratio by 0.2 percentage points and wages by 0.42%. The effects are more pronounced for men, particularly in the manufacturing sector, and vary according to the level of education. Less educated workers experience wage and job cuts, while highly skilled workers receive raises. The study uses regression models, Bartik's approach, and IV estimates to ensure the robustness of the results, confirming the original findings on the negative impact of automation on the labor market.

Contents

Introduction

Rapid advances in technology, especially the rise of industrial robots, have generated a lot of interest and concern about their impact on employment and wages in the United States. This topic is crucial not only for economists and policymakers, but also for the general public, as it touches on the future of work, job security and economic inequality. People outside the field may find this important as the implications of automation could affect their livelihoods and the economy in general, which could lead to discussions about the need for reskilling and education in a changing labour market.

This manuscript addresses the question of how the adoption of industrial robots affects employment and wages in U.S. labor markets. By replicating the study conducted by Acemoglu and Restrepo (2017) and deepening the results obtained, this research aims to better understand the causal relationships between robot adoption and labour market outcomes.

The manuscript makes several important contributions to the existing literature. First, it provides empirical evidence of the effects of industrial robots on job losses and wage variations, filling a gap in the current understanding of the impact of automation. Second, it uses a robust methodological approach using two-step least squares (2SLS) to address potential endogeneity issues, thereby improving the reliability of the results. Third, by focusing on specific sectors and regions, the study provides nuanced insights into how different domains are affected by robot adoption, which is essential for targeted policy interventions.

The results reveal that the adoption of industrial robots has significant effects on employment and wages, contributing to job losses in some sectors while highlighting the potential for wage polarization. These findings are novel as they provide an empirical basis for ongoing debates on the role of automation in labour market dynamics, suggesting that policymakers need to consider these impacts when designing labour market interventions.

The rest of the manuscript is organized as follows: it begins with a replication of the studies already carried out by the authors, followed by a deepening according to genre and level of education. The results section presents the findings, which are then discussed in terms of their implications for future policy and research.

Replication of the study carried out by Acemoglu and Restrepo

The study assigned by these researchers focuses on the impact of industrial robots on the labor market in the United States. Replication parking is available on *MIT Economics*.

1.1 Empirical approach

1.1.1 Data

For this replication, the data used is the same as that used by Acemoglu and Restrepo, including:

- **Industrial Robot Data**

  The researchers are using data from the International Federation of Robotics (IFR) on inventories of industrial robots in different sectors in the United States.

  This data includes information on the use of robots in various manufacturing industries, such as textiles, electronics, and automotive, grouped under the term "IFR industries." The data is limited to some European countries and the United States, with information available for 13 disaggregated industries in manufacturing. Although limitations exist, unclassified robots are assigned to industries according to the proportions in the classified data. Noise in robot exposure measurements is processed using an instrumental variable procedure. Robot penetration is measured over specific time periods, and long-difference models cover the period from 1993 to 2007, with alternative models for other time periods. The analysis of technologically advanced countries in relation to the United States makes it possible to distinguish the impacts of global advances from factors specific to the United States.

  The researchers used various datasets to analyze the impact of robot adoption on industry-level economic factors in the United States. Sources include County Business Patterns (CBPs), providing information on employment, payroll, value added, and labor share. In addition, data on value added and labour shares were obtained from the Bureau of Economic Analysis' (BEA-IO) input-output tables, while information on computer capital and total capital stock came from the Bureau of Labor Statistics. This analysis aims to understand how robot adoption across industries affects employment, wages, and other economic indicators, thereby providing insights into the relationship between robots and labor markets in the United States.

- **Economic data**

  Data from the U.S. Census Bureau and the American Community Survey (ACS) are used to obtain information on employment, wages, and commuting zone demographics.

  The study focuses on 722 transportation areas in the United States to analyze the impact of industrial robots on employment and wages. Researchers measure exposure to robots in each transport area, using this measure to assess local effects on employment and wages. The study focuses on transportation areas as units of analysis covering the entire continental United States. To avoid endogenous and correlated biases in the exposure variable, the reference employment shares are kept constant, even when analyzing changes over sub-periods. Data from the Census and the American Community Survey (ACS) are used to calculate average hourly and weekly wages in 250 demographic cells in the areas, adjusting for changes in worker characteristics.

1.1.2 Regression Models

The regression models used by researchers estimating the impact of robot exposure on employment and wages:

$$d \ln \text{Emploi}_c = \beta_{\text{Emploi}} \cdot \text{Exposure\_to\_Robots}_c + \epsilon_c^{\text{Emploi}} \tag{1.1}$$

$$d \ln \text{Wages}_c = \beta_{\text{Wages}} \cdot \text{Exposure\_to\_Robots}_c + \epsilon_c^{\text{Wages}} \tag{1.2}$$

Where :

- **Emploi** represents the employment variable in each commuting zone.
- **Wages** represents the salary variable in each commuting zone.
- $\epsilon_c^{\text{Emploi}}$ and $\epsilon_c^{\text{Wages}}$ represent the unobserved factors that may influence the results analyzed (changes in local economic conditions, demographic changes,...)

Exposure to Robots

It is measured by the number of industrial robots by sector and by area.

$$\text{Exposure\_to\_Robots}_c = \sum_{i \in I} \ell_{ci} \cdot APR_i \tag{1.3}$$

- ℓ_{ci} represents the industry's share i in total employment in commuting zone c.
- APR_i is the adjusted penetration of robots in industry i in the United States. It is defined by

$$APR_i = \frac{d\theta_i}{1 - \theta_i} \frac{\gamma_L}{\gamma_M} \tag{1.4}$$

with

- θ_i advancement in automation technology.
- γ_L labour productivity.
- γ_M robot productivity.

Regression model assumptions

The validity of the models is based on three critical assumptions.

- First, the authors assume that the variable, exposure to robots, is exogenous and is not correlated with the error term in the employment and wage equations, thus ensuring unbiased estimates.
- Second, this variable must be relevant, i.e., it must significantly influence robot exposure, which the authors demonstrate through the relationship between industry-level robot adoption and local labor market outcomes.
- Finally, they hypothesize that the areas most exposed to robots after 1990 do not show differential trends in employment and wages before this period, confirming that the observed effects are not due to pre-existing trends.

1.1.3 Instrumentation

Bartik's approach

To address the problems of correlation between robot adoption and pre-existing economic trends, the researchers use Bartik's approach, which instrumentises robot exposure with robot adoption trends in more advanced European robotics countries (Germany, Japan).

The Bartik approach combines industry-level variations in robot use with benchmark employment shares. This approach is used by researchers to isolate the impact of technological change on robot penetration (potential correlations between U.S. exposure to robots and error terms).

Exposure to robots is calculated by the following formula:

$$\text{Exposure\_to\_Robots}_c = \sum_{i \in I} \ell_{ci} \cdot \overline{APR}_i, \tag{1.5}$$

with $\overline{APR}_i$ the adjusted penetration of robots calculated from data from European countries.

Bartik's approach assumptions

The researchers propose several hypotheses to solve the problems of correlation between robot adoption and pre-existing economic trends using Bartik's approach.

- First, they hypothesize that the instrument, i.e., trends in robot adoption in frontier countries such as Germany and Japan, is exogenous, meaning that it should not be correlated with error terms in their employment and wage equations.
- Second, they argue that this instrument is relevant, as it significantly influences the U.S. exposure to robots due to industry-level variations in robot use combined with benchmark employment shares.
- Finally, they believe that the Bartik approach effectively isolates the impact of technological change on robot penetration, thereby mitigating potential correlations between exposure to U.S. robots and error terms, thus ensuring that the observed effects on employment and wages are attributable to robot adoption rather than confounding factors.

IV Estimates

IV Estimates are used to address situations in which an explanatory variable is correlated with the error term, resulting in biased and inconsistent estimates in ordinary least squares (OLS) regression. It uses instruments, i.e., variables correlated with the endogenous explanatory variable but not correlated with the error term, to provide consistent estimates. The most commonly used method is Two-step Least Squares methods.

Two-step Least Squares (2SLS):

The paper applies the 2SLS method to estimate the effects of robot adoption on employment and wages. First, the authors regress the endogenous variable (e.g., employment or wages) of the instrument (exposure to robots) to obtain predicted values. These predicted values are then used in the second step of the analysis to estimate the causal impact of robots on labor market outcomes.

IV Estimates assumptions

The main assumption of IV Estimates is that the instrument affects the dependent variable only through its effect on the endogenous explanatory variable. This means that the instrument must meet two key conditions:

- **Relevance :** the instrument must be correlated with the endogenous variable.
- **Exogeneity :** the instrument must not be correlated with the error term in the regression equation.

1.2 Results

1.2.1 Exposure of robots vs. integrators

The study shows a positive association between robot exposure and the number of integrators, suggesting that areas with higher robot penetration are indeed adopting more industrial robots. The following residual graph illustrates the change in the log of one plus the number of integrators in a path area compared to the exposure to robots. The dotted line of the graph represents the regression relationship after excluding the 1 % of the path areas most exposed to robots, thus reinforcing the robustness of the positive correlation observed.

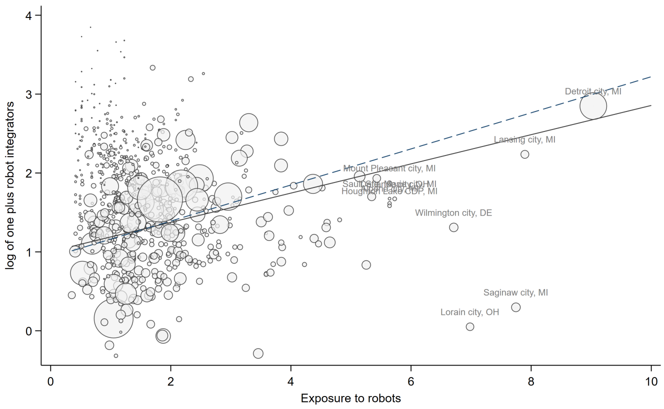


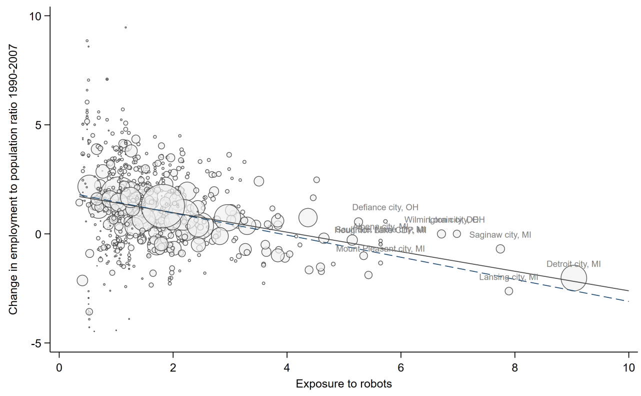
Figure 1.1: Log change of one plus the number of integrators in a commuting zone versus exposure to robots

1.2.2 Effects of Robot Exposure on Employment and Wages

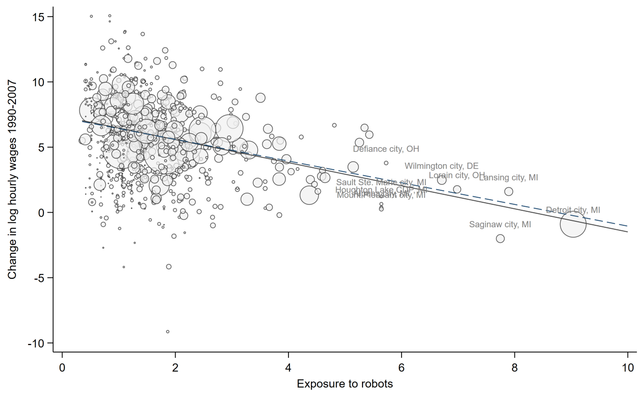
Significant advances in robot technology in the 1990s and 2000s led to increased adoption of robots, with an empirical approach based on a model where robots and workers compete in the production of tasks. This adoption has negative effects on wages and employment due to displacement and productivity effects.

The researchers find that an increase of one robot per thousand workers reduces the employment-to-population ratio by 0.2 percentage points. This shows that as robots are adopted, they are replacing human workers, leading to a decline in the job supply.

In addition to the impacts on employment, the study indicates that the same increase in robot adoption is correlated with a 0.42% decrease in wages. This shows how automation affects not only the number of jobs, but also the compensation of the remaining workers.



Change in the employment-to-population ratio by exposure to robots.



Change in log hourly wages by exposure to robots.

Figure 1.2: Relationship between employment, hourly wages and exposure to robots.

1.2.3 The effects of robots on employment and wages by gender

The impact is negative for both men and women, but more pronounced for the former, with a decline in employment of 0.57% for men and 0.34% for women per population over the period 1990-2007 (long-term estimate in Table 1). Male employment losses are more concentrated in the manufacturing sector, while women's employment losses occur mostly in non-manufacturing sectors.

In the long-term estimates (columns 1 and 4 Panel B of Table 1), increased exposure to robots is associated with a decrease in hourly wages (log) for both men and women, with an overall stronger impact for men. There is a decrease of around 0.98% for men, compared to about 0.79% for women per population.

Table 1: *The effects of robots on employment and wages by gender*

	Long differences						Stacked differences					
	Men			Women			Men			Women		
	Weighted by population	Excludes zones with highest exposure	Unweighted	Weighted by population	Excludes zones with highest exposure	Unweighted	Weighted by population	Excludes zones with highest exposure	Unweighted	Weighted by population	Excludes zones with highest exposure	Unweighted
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A. Change in the employment to population ratio												
Exposure to robots	-0.567	-0.674	-0.652	-0.336	-0.473	-0.385	-0.684	-0.809	-0.972	-0.418	-0.590	-0.515
	(0.065)	(0.147)	(0.148)	(0.063)	(0.145)	(0.112)	(0.065)	(0.175)	(0.134)	(0.049)	(0.133)	(0.079)
Observations	722	712	722	722	712	722	1444	1424	1444	1444	1424	1444
R-squared	0.62	0.59	0.59	0.71	0.71	0.58	0.37	0.35	0.37	0.45	0.45	0.37
Panel B. Change in log of hourly wages												
Exposure to robots	-0.979	-0.973	-0.984	-0.787	-0.591	-0.879	-1.651	-1.926	-1.914	-1.216	-1.343	-1.431
	(0.148)	(0.291)	(0.241)	(0.137)	(0.306)	(0.223)	(0.211)	(0.601)	(0.320)	(0.158)	(0.512)	(0.314)
Observations	43599	42935	43599	43501	42841	43501	92008	90612	92008	91598	90206	91598
R-squared	0.29	0.28	0.07	0.34	0.33	0.08	0.25	0.24	0.08	0.31	0.30	0.11
Panel C. Change in manufacturing employment to population ratio												
Exposure to robots	-0.256	-0.386	-0.442	-0.068	-0.094	-0.238	-0.314	-0.420	-0.507	-0.137	-0.224	-0.338
	(0.064)	(0.137)	(0.112)	(0.029)	(0.061)	(0.062)	(0.037)	(0.092)	(0.105)	(0.027)	(0.064)	(0.060)
Observations	722	712	722	722	712	722	1444	1424	1444	1444	1424	1444
R-squared	0.70	0.70	0.69	0.85	0.85	0.83	0.49	0.48	0.52	0.67	0.67	0.67
Panel D. Change in non-participation rate												
Exposure to robots	0.364	0.276	0.398	0.201	0.205	0.263	0.658	0.951	0.850	0.318	0.578	0.375
	(0.063)	(0.119)	(0.129)	(0.036)	(0.144)	(0.112)	(0.096)	(0.198)	(0.117)	(0.082)	(0.188)	(0.095)
Observations	722	712	722	722	712	722	1444	1424	1444	1444	1424	1444
R-squared	0.51	0.48	0.40	0.71	0.72	0.58	0.55	0.55	0.37	0.43	0.43	0.34
Covariates:												
Baseline covariates	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

The table presents estimates of exposure to robots, as well as their effects on employment, wages and other indicators, by gender. Columns 1 to 6 show long-term estimates for the period 1990-2007, while columns 7 to 12 present stacked estimates for the periods 1990-2000 and 2000-2007. Regressions are performed at the level of commuting areas, with demographic cells defined by age, gender, education level and ethnicity. Columns 1-2, 4-5, 7-8 and 10-11 correspond to 1990 population-weighted regressions. Columns 2, 5, 8 and 11 exclude the commuting areas most exposed to robots (the top percentile). Columns 3, 6, 9 and 12 present estimates from unweighted regressions.

All specifications include:

- *dumb variables for census divisions,*
- *temporal indicators in stacked regressions,*
- *the socio-demographic characteristics of commuting areas (log of the total population, share of women, share of the population aged over 65, share of the population by level of education: without a degree, university degree, college, vocational, master's and doctorate),*
- *shares of different ethnic origins (White, Black, Hispanic, Asian),*
- *the share of employment in manufacturing and light industry,*
- *women's share of manufacturing employment,*
- *exposure to imports from China,*
- *and the share of routine jobs.*

Standard errors robust to heteroscedasticity and intrastate correlation are shown in parentheses.

1.2.4 Effects of robots on employment and wages: commuting zone level

According to the estimates presented in columns 3 and 6 of Part A (Table 2), the addition of one additional robot per thousand workers in a commuting area decreases its employment-to-population ratio by 0.39 percentage points and its hourly wage by 0.77% compared to other commuting areas. This data indicates that one additional robot results in the loss of about six jobs in the affected area, compared to the others. These results are tested reliably using the IV estimates's method.

Table 2: *IV Estimates of exposure to robots, employment and wages during different time periods*

	Change in the employment to population ratio			Change in log hourly wages		
	(1)	(2)	(3)	(4)	(5)	(6)
			Panel A. Long-differences, 1990-2007			
US exposure to robots	-0.375	-0.377	-0.388	-1.022	-0.762	-0.768
	(0.116)	(0.121)	(0.125)	(0.269)	(0.237)	(0.238)
Observations	19	19	19	19	19	19
			Panel B. Alternative imputation of US data, 1990-2007			
US exposure to robots	-0.388	-0.391	-0.402	-1.060	-0.790	-0.796
	(0.120)	(0.126)	(0.130)	(0.279)	(0.246)	(0.246)
Observations	19	19	19	19	19	19
			Panel C. Long-differences, 1990-2014			
US exposure to robots	-0.303	-0.241	-0.250	-1.268	-1.103	-1.128
	(0.111)	(0.055)	(0.062)	(0.129)	(0.113)	(0.127)
Observations	19	19	19	19	19	19
			Panel D. Long-differences, 2000-2007			
US exposure to robots	-0.623	-0.574	-0.585	-1.376	-1.147	-1.191
	(0.084)	(0.055)	(0.063)	(0.240)	(0.064)	(0.069)
Observations	19	19	19	19	19	19
			Panel E. Long-differences, 2000-2014			
US exposure to robots	-0.451	-0.327	-0.339	-1.590	-1.566	-1.601
	(0.147)	(0.021)	(0.023)	(0.078)	(0.052)	(0.054)
Observations	19	19	19	19	19	19
Covariates:						
Division dummies	✓	✓	✓	✓	✓	✓
Demographics and industry shares		✓	✓		✓	✓
Trade, routine jobs			✓			✓

The table presents IV Estimates of exposure to robots, employment and wages during different time periods. In all models, we instrument the U.S. robot exposure using EURO5 robot exposure. Robust standard errors calculated according to Borusyak et al. (2018) and are shown in parentheses.

1.2.5 Effects of robots on industries

Over the past 14 years (1993 - 2007), there has been a correlated increase in robot adoption between EURO5 countries and the United States. Nevertheless, the extent of this adoption varies greatly from industry to industry. For example, automotive, plastics and chemicals, and metallurgy saw increases of more than 7.5 robots per thousand workers, while sectors such as paper and printing, textiles, wood and furniture saw only modest growth (Figure 1.4).

The study shows that the effects of robots are concentrated in the manufacturing industry (Figure 1.3 and also illustrated in Table 1) and in particular in highly robotized industries, including automotive, plastics and chemicals, metal products, base metals, electronics, and food and beverage. There are no significant effects on the rest of the manufacturing industries. The approach used here is Bartik's.

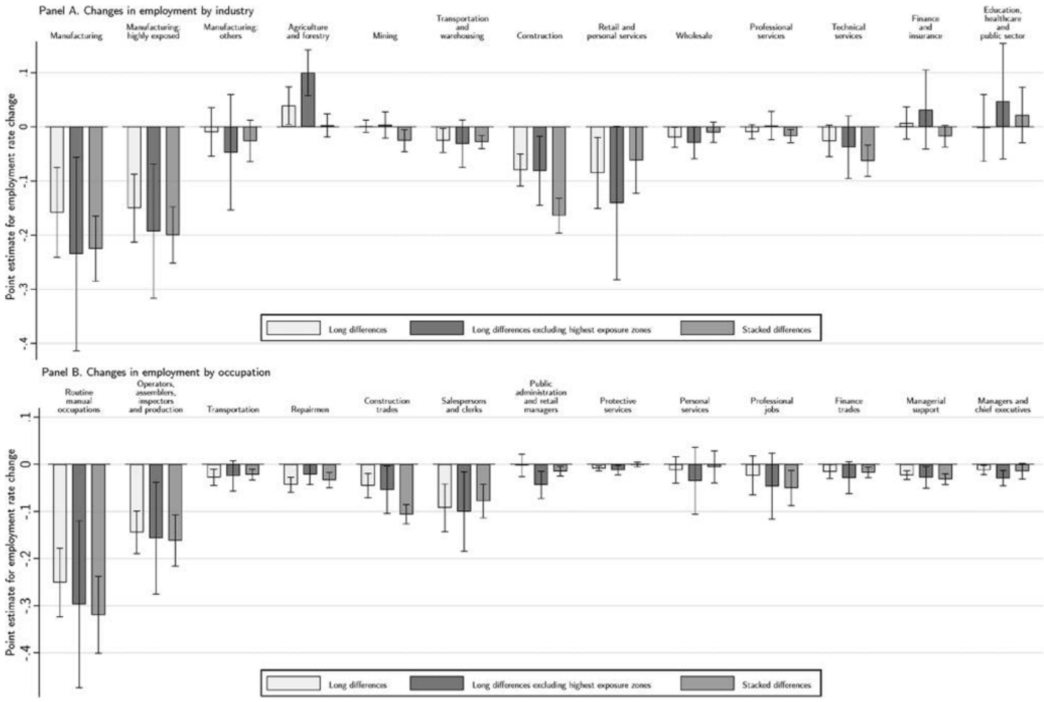


Figure 1.3: Effects of robots on industries and professions.

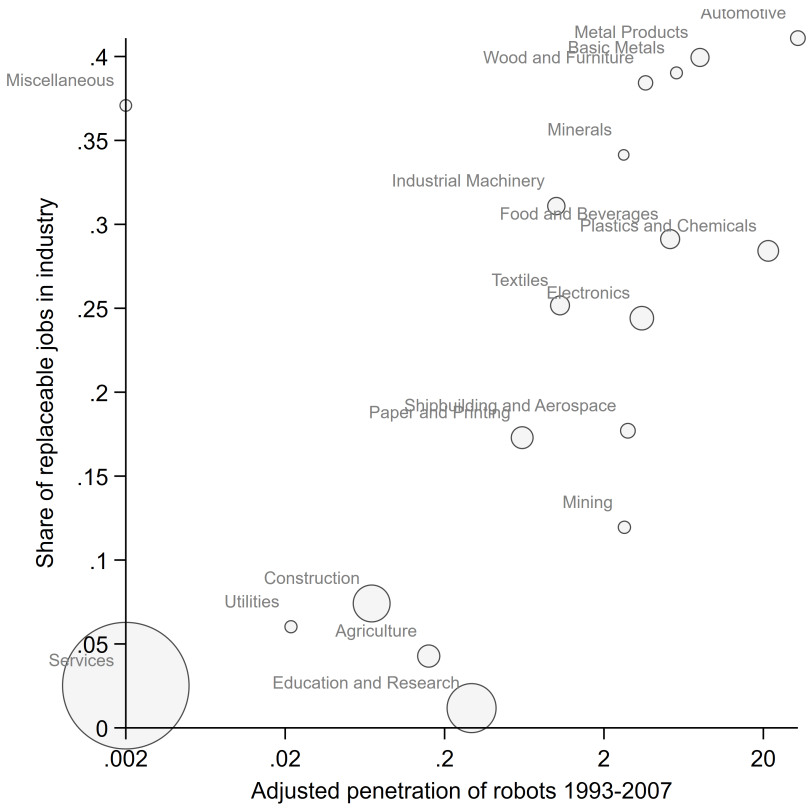


Figure 1.4: Adjusted robot penetration in the US and EURO5 by industry.

1.2.6 Effects of robots on employment and wages by education and gender

Robots have negative effects on employment and wages for men and women with low or medium levels of education (no high school diploma, high school diploma, college or vocational education). Surprisingly, no positive effect is noted for master's or doctoral graduates, which can be explained by a lower demand for these profiles in the non-profit sector or by the fact that industrial robots do not directly complement highly skilled workers.

As far as wages are concerned, the negative effects are mainly concentrated at low and medium levels of remuneration. For all employees, there were significant declines below the 35^{e} percentile of the wage distribution. For workers without a university degree, these negative effects extend up to the 85^{e} percentile, while for university graduates or above, they are limited to wages below the 15^{e} percentile.

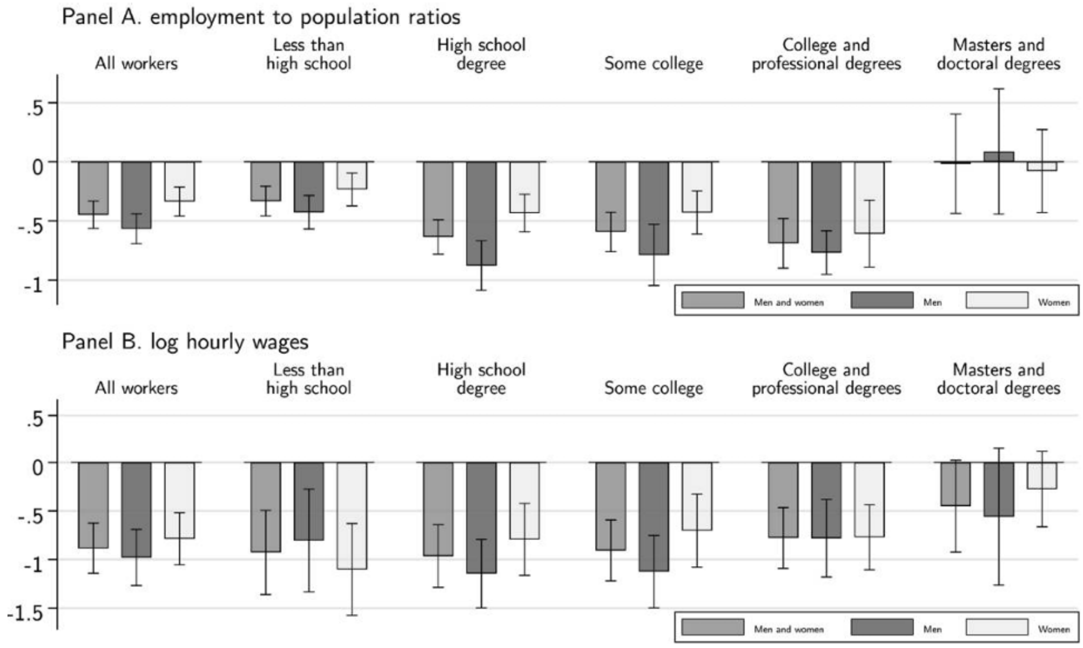
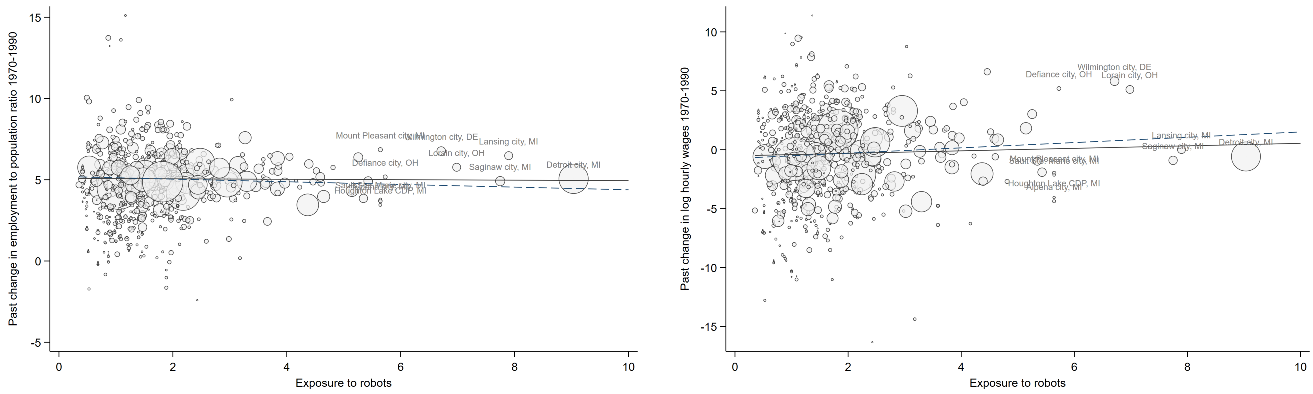


Figure 1.5: Effects of exposure to robots on employment and wages by education and gender

1.3 Pretrends analysis

To ensure that they had properly measured the effect of robot adoption over the study period, the researchers examined the predictable trends in labour market outcomes prior to the mass adoption of robots. The following graphs show that the areas most exposed to robots did not show differential trends before 1990. This lack of pretensions supports the argument that the impact of robots is distinct from that of other financial and technological changes, thus reinforcing the validity of the results.



Relation entre l'exposition aux robots et rapport emploi/population pour la période 1970-1990.

Relationship between exposure to robots and the employment/population ratio for the period 1970-1990.

Figure 1.6: Employment and wage trends

1.4 Robustness Test Analysis

The study uses various robustness tests to validate its findings regarding the impact of robots on employment and wages. An important approach is the variation of the measurement of exposure to robots. The authors test different methods, including calculating exposure on the basis of averages from various European countries and using different distributions of employment between 1990 and 1970. This ensures that the construction of the exhibition does not significantly influence the results

Another essential aspect of robustness checks concerns the control of international competition. The analysis incorporates controls on exposure to imports from Mexico, offshoring of tasks, and exports from countries with advanced robotics, such as Germany, Japan, and South Korea. By addressing these potential confounding factors, the study reinforces its argument that the observed impacts on employment and wages are indeed attributable to the adoption of robots rather than other external influences. In addition, the inclusion of fixed government effects allows for unobserved heterogeneity across states, further isolating the effects of robots on labour market outcomes (Tables 1 and 2).

Our contribution

2.1 Effects of robots on employment and wages: a polynomial regression model of degree 2

The authors carried out their studies using a linear regression model of logarithm of employment and wages on robot exposure and other specification variables such as dummy variables for census divisions, temporal indicators, socio-demographic characteristics, etc.

In this session, we will perform a polynomial regression by introducing the square of the variable exposure to robots. We are faced with the following model:

$$\begin{aligned} d\ln \text{Emploi}_c = {} & \beta_1 \cdot \text{Exposure\_to\_Robots}_c + \beta_2 \cdot \text{Exposure\_to\_Robots}_c^2 + \beta_3 \cdot \text{Demographics}_c \\ & + \beta_4 \cdot \text{Industry\_shares} + \beta_5 \cdot \text{Region\_effects} + \beta_6 \cdot \text{Gender} + \beta_7 \cdot \text{Other\_shocks} + \epsilon_c^{\text{Emploi}} \end{aligned} \tag{2.1}$$

$$\begin{aligned} d\ln \text{Wages}_c = {} & \alpha_1 \cdot \text{Exposure\_to\_Robots}_c + \alpha_2 \cdot \text{Exposure\_to\_Robots}_c^2 + \alpha_3 \cdot \text{Demographics}_c \\ & + \alpha_4 \cdot \text{Industry\_shares} + \alpha_5 \cdot \text{Region\_effects} + \alpha_6 \cdot \text{Gender} + \alpha_7 \cdot \text{Other\_shocks} + \epsilon_c^{\text{Wages}} \end{aligned} \tag{2.2}$$

with Gender a variable that takes 1 if it is women.

Table 3: *Effects of robot exposure on employment and wages by gender: polynomial regression model of degree 2*

	Long differences						Stacked differences					
	Men			Women			Men			Women		
	Weighted by population	Equally sized with highest exposure	Unweighted	Weighted by population	Equally sized with highest exposure	Unweighted	Weighted by population	Equally sized with highest exposure	Unweighted	Weighted by population	Equally sized with highest exposure	Unweighted
	Panel A : Change in the employment-to-population ratio											
Exposure to robots	-0.774	-1.079	-1.169	-0.631	-0.738	-0.627	-0.754	-0.451	17.608	-0.652	-0.340	-0.818
	(0.203)	(0.363)	(0.395)	(0.197)	(0.318)	(0.294)	(0.253)	(0.328)	(21.449)	(0.183)	(0.251)	(0.205)
Observations	2.4e+08	712	722	722	712	722	1444	1424	92009	1444	1424	1444
R-squared	0.62	0.59	0.59	0.71	0.71	0.58	0.37	0.35	0.04	0.46	0.45	0.37
	Panel B : Change in log of weekly wages											
Exposure to robots	-1.177	-2.277	-1.381	-0.425	-0.547	-1.019	-1.888	-1.103	-2.476	-1.135	0.066	-1.909
	(0.467)	(0.818)	(0.719)	(0.482)	(0.819)	(0.576)	(0.906)	(1.281)	(0.836)	(0.772)	(1.170)	(0.729)
Observations	60519952	58682649	4.0e+10	55096333	53432458	3.6e+10	1.2e+08	1.2e+08	1.6e+11	1.1e+08	1.1e+08	1.4e+11
R-squared	0.29	0.28	0.07	0.34	0.33	0.08	0.25	0.24	0.08	0.31	0.30	0.11
	Panel C : Change in manufacturing employment to population											
Exposure to robots	-0.564	-0.517	-0.589	-0.185	-0.209	-0.418	-0.470	-0.295	17.608	-0.276	-0.102	-0.596
	(0.175)	(0.251)	(0.198)	(0.093)	(0.134)	(0.133)	(0.117)	(0.169)	(21.449)	(0.096)	(0.114)	(0.108)
Observations	2.4e+08	712	722	722	712	722	1444	1424	92009	1444	1424	1444
R-squared	0.71	0.70	0.69	0.85	0.85	0.83	0.49	0.48	0.04	0.67	0.67	0.67
	Panel D : Change in non-participation rate											
Exposure to robots	0.249	0.555	0.514	0.248	0.037	0.114	1.001	0.597	17.608	0.701	0.091	0.553
	(0.180)	(0.223)	(0.320)	(0.228)	(0.335)	(0.352)	(0.287)	(0.349)	(21.449)	(0.290)	(0.357)	(0.243)
Observations	2.4e+08	712	722	722	712	722	1444	1424	92009	1444	1424	1444
R-squared	0.51	0.48	0.40	0.71	0.72	0.58	0.55	0.55	0.04	0.43	0.43	0.34
Baseline covariates	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

The empirical analysis highlights significant effects of automation — as measured by exposure to robots — on the labour market, for both men and women. Focusing on the results in columns 1 and 4 for long differences, and columns 7 and 10 for stacked differences, the following observations can be made.

2.1.1 Impact on employment by gender

In Panel A, exposure to robots is associated with a significant decline in the employment rate. For men, an increase in exposure results in a decrease of 0.774 percentage points in the employment rate in the long-difference data, and a 0.754-point decrease in the stacked differences.

For women, the effect is also negative: the decline is estimated to be 0.631 percentage points in the long differences, and as high as 0.652 percentage points in the stacked differences. This suggests that automation negatively affects the employment of both genders, with a greater impact among women in the stacked data, which could reflect long-term cumulative effects.

2.1.2 Impact on wages by gender

In Panel B, there is also a significant negative effect on weekly wages. In men, exposure to robots reduces wages by 1.18 % in long differences, and by 1.88 % in stacked differences.

For women, wages decrease by 0.42 % in the long differences, and by 1.13 % in the stacked data. The effect is therefore present for both groups, with a relatively greater impact in men, especially in the case of stacked differences.

In summary, women seem to be more affected in terms of employment, with a sharper decline in the employment rate, especially in the stacked data. This suggests a higher vulnerability to labour market exclusion in the long term. Men, on the other hand, are experiencing a greater deterioration in wages and manufacturing employment, sectors in which they are historically more represented. They also show a larger increase in non-participation, which may reflect a permanent exit from the labour market.

2.1.3 Impact on employment by gender and education

As we have just seen, to study the impact on education, we consider the model by introducing variable variables indicating education levels. However, we will consider the employment-to-population ratio as the dependent variable for employment. The bar graph (Figure 2.1) is obtained.

Overall, the results indicate that exposure to robots has a significant negative effect on employment. This impact is particularly marked among men, compared to women, and tends to increase with the level of education.

2.1.4 Impact on wages by gender and education

Regarding wages (Panel B), robotization also leads to a general decrease in hourly wages, measured here in logarithms. However, this decline is more pronounced among men, and particularly among those with a higher level of education. Conversely, women with degrees, particularly at the master's and doctoral levels, seem to be less affected, and sometimes even slightly favoured. This differentiation may be linked to the gendered structure of jobs, as women are over-represented in sectors less exposed to automation, such as care, education or social services.

In summary, the graph highlights that the impact of robots is neither neutral nor uniform: it depends strongly on the level of education and gender. These results underline the need for a targeted public policy approach, in particular to anticipate the risks of occupational exclusion in certain categories of workers, and to adapt training and retraining systems.

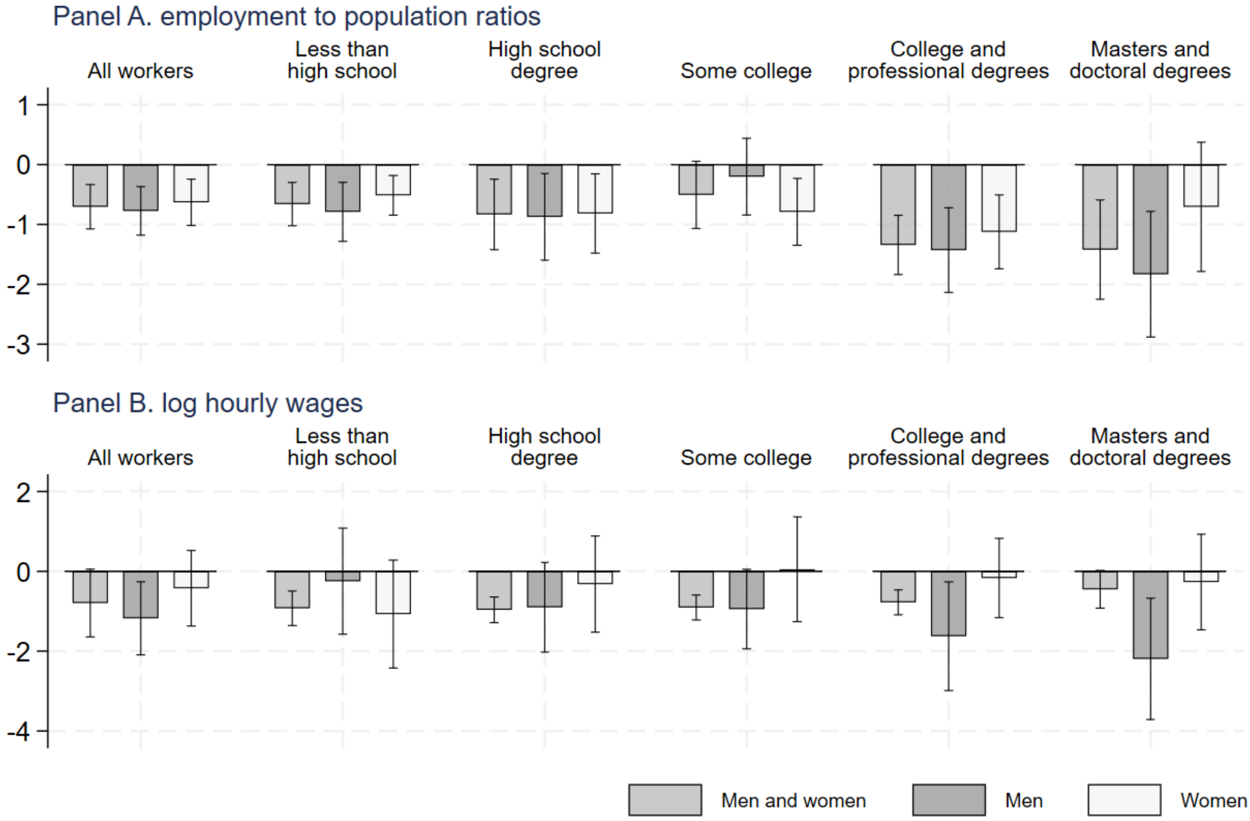


Figure 2.1: Effects of robot exposure on employment and wages by gender and education: polynomial regression model of degree 2

2.2 Effects of robots on employment and wages:a linear regression model including interaction variables

In this session, we will introduce the interaction variables into the model.

$$\begin{aligned} d\ln \text{Emploi}_c = {} & \beta_1 \cdot \text{Exposure\_to\_Robots}_c + \beta_2 \cdot \text{Exposure\_to\_Robots}_c * \text{interaction} \\ & + \beta_3 \cdot \text{Demographics}_c + \beta_4 \cdot \text{Industry\_shares} + \beta_5 \cdot \text{Region\_effects} + \beta_6 \cdot \text{Gender} \\ & + \beta_7 \cdot \text{Other\_shocks} + \epsilon_c^{\text{Emploi}} \end{aligned} \tag{2.3}$$

$$\begin{aligned} d\ln \text{Wages}_c = {} & \alpha_1 \cdot \text{Exposure\_to\_Robots}_c + \alpha_2 \cdot \text{Exposure\_to\_Robots}_c * \text{interaction} \\ & + \alpha_3 \cdot \text{Demographics}_c + \alpha_4 \cdot \text{Industry\_shares} + \alpha_5 \cdot \text{Region\_effects} + \alpha_6 \cdot \text{Gender}+ \\ & + \alpha_7 \cdot \text{Other\_shocks} + \epsilon_c^{\text{Wages}} \end{aligned} \tag{2.4}$$

First, we will introduce the interaction variable that takes 1 if the age is above 65 years old (ipums_above65). This variable will allow us to analyze whether the effect is the same for young and old (65 years or older).

Table 4: *Effects of Exposure to Robot on employment and wages with a linear regression model including ipums_above65-Interaction variables*

	Long differences						Stacked differences					
	Men			Women			Men			Women		
	Weighted by population	Equally sized with highest exposure	Unweighted	Weighted by population	Equally sized with highest exposure	Unweighted	Weighted by population	Equally sized with highest exposure	Unweighted	Weighted by population	Equally sized with highest exposure	Unweighted
Panel A : Change in the employment-to-population ratio												
Exposure to robots	-1.066	-1.374	-0.348	-0.969	-1.572	-0.119	-0.341	-0.221	-0.696	0.001	-0.249	0.177
	(0.271)	(0.482)	(0.640)	(0.318)	(0.471)	(0.447)	(0.810)	(1.256)	(0.654)	(0.648)	(0.974)	(0.546)
Observations	2.4e+08	712	722	722	712	722	1444	1424	1444	1444	1424	1444
R-squared	0.62	0.59	0.59	0.71	0.72	0.58	0.37	0.31	0.31	0.41	0.41	0.31
Panel B : Change in log of weekly wages												
Exposure to robots	-0.416	-0.782	1.011	-0.959	-0.642	-0.370	-1.452	-2.890	0.435	-1.301	-1.547	-0.799
	(0.663)	(1.094)	(1.029)	(0.658)	(0.893)	(0.809)	(1.209)	(1.426)	(1.011)	(1.260)	(1.687)	(0.852)
Observations	43267	42603	43600	43142	42482	43501	91140	89744	92009	90672	89280	91598
R-squared	0.10	0.08	0.04	0.05	0.05	0.02	0.10	0.08	0.04	0.16	0.15	0.06
Panel C : Change in manufacturing employment to population												
Exposure to robots	-0.220	-0.553	-0.372	-0.039	-0.175	0.145	0.143	0.340	-0.185	0.190	0.103	0.156
	(0.230)	(0.404)	(0.287)	(0.109)	(0.132)	(0.154)	(0.219)	(0.397)	(0.335)	(0.146)	(0.204)	(0.215)
Observations	2.4e+08	712	722	722	712	722	1444	1424	1444	1444	1424	1444
R-squared	0.70	0.70	0.69	0.85	0.85	0.83	0.50	0.41	0.42	0.63	0.63	0.62
Panel D : Change in non-participation rate												
Exposure to robots	0.023	-0.508	-0.135	0.579	0.911	-0.284	-1.562	-2.255	-0.353	-1.202	-1.124	-0.983
	(0.333)	(0.318)	(0.598)	(0.337)	(0.521)	(0.424)	(1.083)	(1.551)	(0.568)	(0.825)	(1.251)	(0.631)
Observations	2.4e+08	712	722	722	712	722	1444	1424	1444	1444	1424	1444
R-squared	0.51	0.49	0.41	0.72	0.72	0.58	0.56	0.55	0.35	0.39	0.39	0.28
Baseline covariates	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

2.2.1 Impact on employment by gender and interacting with ipums_above65

The results also have a significant impact on the population-based employment rate for men and women in the general age group. The negative coefficients -1.006 for males and -0.900 for females, with standard deviations of 0.271 and 0.318, respectively, indicate that exposure to robots is followed by a significant decrease in the employment-population ratio. This is in favor of the fact that automation could lead to job loss, which could be achieved by replacing labor agents with machines in certain sectors. But for the oldest workers (ipums_above65), the effects are not statistically significant, which could be explained by the fact that the oldest workers are less affected by these changes or that they receive specific protection measures.

2.2.2 Impact on wages by gender and interacting with ipums_above65

Analysis of significant coefficients reveals that exposure to industrial robots has a notable impact on weekly wages, especially for men in the general population. The positive coefficient of 0.416, with a standard deviation of 0.663, indicates that exposure to robots is associated with a significant increase in weekly wages for men. This raises the question of whether automation will lead to better rewards, possibly due to increased productivity or a demand for special skills. In contrast, for women in the general population, the effect is less pronounced and not significant, which could indicate gender disparities in the wage benefits associated with automation.

2.3 Effects of robots on employment and wages: a model with interaction of education levels

We will examine the interaction between robot exposure and variables related to education levels, including nipums_highschool, nipums_college and nipums_masters. Our objective is to analyze the impact of these interactions on the employment/population ratio. The following diagram illustrates the results obtained.

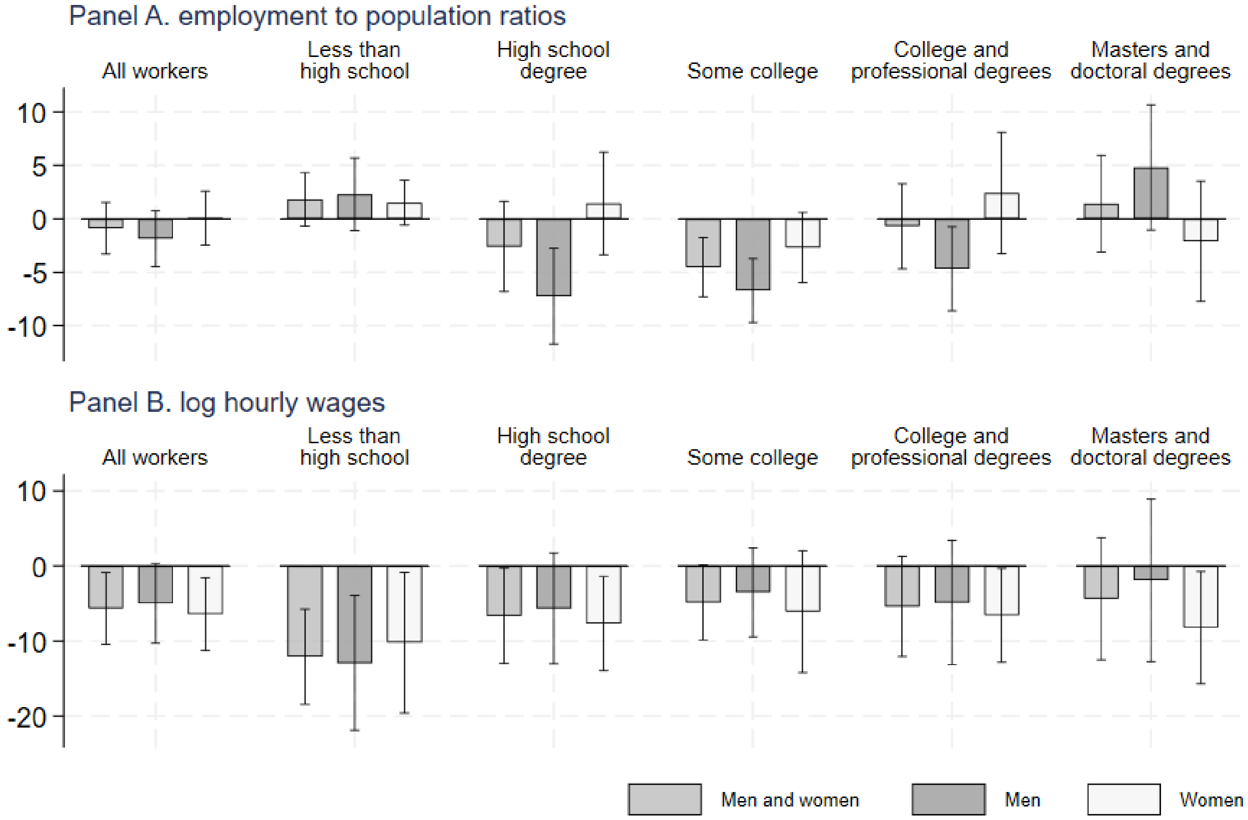


Figure 2.2: Effects of Exposure to Robot on employment and wages with a linear regression model including interaction of education levels

2.3.1 Impact on employment by gender and interaction of education levels

An examination of the population-based employment rate shows that exposure to robots has varied effects depending on the level of education and gender. Workers with a few years of college or a college and vocational degree see a reduction in their population-based employment rate, with this effect being more pronounced among men. Conversely, highly skilled workers, especially those with a master's or doctoral degree, benefit from an increase in their employment-to-population ratio, with particularly strong positive effects for men. These findings indicate that automation can lead to job losses for middle-skilled workers, while creating opportunities for the most educated workers.

2.3.2 Impact on wages by gender and interaction of education levels

The diagram reveals that exposure to robots has varying effects on hourly wages depending on education level and gender. Less educated workers, especially those with less than a high school diploma, experience a significant reduction in their hourly wages, with this effect being more pronounced among men. In contrast, highly skilled workers, including those with master's or doctoral degrees, benefit from an increase in their hourly wages, with particularly strong positive effects for men. Automation then tends to amplify wage inequalities, favouring the most qualified workers while penalising the least educated.

Conclusion

This replication study of the work of Acemoglu and Restrepo (2017) confirms the significant impact of industrial robots on labor markets in the United States. The results show that the increasing adoption of robots has negative effects on employment and wages, with notable disparities across gender, education level and industries. Lower-skilled workers, especially men, are the most affected, suffering job losses and significant wage cuts. In contrast, highly skilled workers, especially those with advanced degrees, enjoy better job opportunities and higher wages.

These findings highlight the need for proactive public policy to manage technological transformations and mitigate their negative effects on vulnerable workers. Policymakers should consider targeted measures, such as vocational training, reskilling and support policies for the most affected sectors. In the age of increasing automation, it is essential to prepare the workforce for these changes and ensure a just transition for all workers.

Bibliography

[1] Acemoglu, D., & Restrepo, P. (2017). *Robots and Jobs: Evidence from US Labor Markets*. Journal of Political Economy.